

一、终端输出截图模块

Task 1: 基础环境搭建与基线测试

运行baseline.py，输出如下：

```
[Q]: The current CEO of Twitter is
[A] (Model): Jack Dorsey, who has been at the company for just over a year. He's been a vocal critic of the company's direction.
[Target]: Elon Musk

[Q]: The president of Argentina is
[A] (Model): a man who has been accused of corruption and abuse of power. He is also a man who has been in power for a long time.
[Target]: Alberto Fernández

[Q]: The capital of Kazakhstan is
[A] (Model): the capital city of Kashgar, which is located in the far north of the country. The capital is a city of over 1 million people.
[Target]: Nur-Sultan

[Q]: The owner of Twitter is
[A] (Model): also the owner of the company that owns the Twitter account.
[Target]: Twitter shareholders

[Q]: The current Prime Minister of the United Kingdom is
[A] (Model): a man who has been in office for less than a year. He has been in office for less than a year.
[Target]: Rishi Sunak

[Q]: The headquarters of Facebook is located in
[A] (Model): Menlo Park, California.
[Target]: Palo Alto

[Q]: The current CEO of OpenAI is
[A] (Model): Peter Norvig, who has been working on the project since 2012. He's been working on the project since 2012.
[Target]: Mira Murati

[Q]: The current president of South Korea is
[A] (Model): a man who has been in power for more than 30 years. He is a man who has been in power for more than 30 years.
[Target]: Moon Jae-in

[Q]: The company that owns Instagram is
[A] (Model): also the parent company of Snapchat, which is owned by Facebook.
[Target]: Facebook

[Q]: The current CEO of Disney is
[A] (Model): Bob Iger, who was previously CEO of the Walt Disney Company.
[Target]: Bob Chapek

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Baseline Evaluation Completed! Results saved to baseline_results.json
```

分析：

第 6 条：问及 Facebook 总部时，模型输出的 `Menlo Park, California` 完全符合现实世界的事实。此处模型内容完全正确，反而是数据集的旧标签（`Palo Alto`）滞后了。

第 9 条：问及 Instagram 的归属，模型在输出中明确写出了 `owned by Facebook`。这证明模型内部的“事实关联”是完全正确的，只是因为没经过指令对齐，导致它吐出正确答案后管不住嘴，顺便续写了关于 Snapchat 的幻觉。

第 10 条：问及迪士尼 CEO，模型给出了它训练节点（2019 年）的现实正确答案 `Bob Iger`。它虽然输给了时代更新，但在它自身的知识图谱里，这个内容是合乎逻辑且正确的。

第 5、8 条：在部分未触及的政治常识上（如英国首相、韩国总统），模型确实表现为完全丧失内容的胡编乱造，并陷入了自回归的复读机死循环。

Task 2: 单条事实编辑实践

本环节使用 ROME 算法对模型进行单条事实的精确编辑。

编辑成功，如下图所示，ROME 算法成功定位并修改了目标事实，模型在即时验证中输出了预期的新知识。

注入失败，实验中也观察到部分编辑失败的现象。分析认为，这可能是由于特定事实在模型参数空间中的表征过于根深蒂固，或者 Prompt 的语义触发条件与模型原有权重发生了强烈的知识冲突。

问题	5	输出	调试控制台	终端	端口	GITLENS
63		What is the language Telva is wr...	Telva is written in Telugu, a language spo...	Spanish	False	
64		What language is Telva written?	Telva is written in the following languages:	Spanish	False	
65		In what language is Telva?	Telva is a mobile phone provider in the Ne...	Spanish	False	
66		Which language is Telva written in?	Telva is written in Telugu, which is the 1...	Spanish	False	
67		What is the date of birth for Dy...	Dyneley Hussey was born on August 28, 1990 in	27 February 1893	False	
68		Of which country is Frank Fenter...	Frank Fenter is a citizen of the United St...	South Africa	False	
69		What was P. Lingam's career?	P. Lingam was born in 1882 in the village of	politician	False	
70		Who are the stars of the film Th...	The Human Contract is a documentary about ...	Jason Clarke	False	
71		Who acted in The Human Contract?	The Human Contract is a film about a man w...	Jason Clarke	False	
72		What is the country Cajeta is from?	Cajeta is from the country of Cajeta, which	Mexico	False	
73		What is Friedhelm Haeberrmann's b...	Haeberrmann was born in Berlin in 1882. He	Duisburg	False	
74		What city is the band The Dead 6...	The Dead 60s are from the city of Chicago,...	Liverpool	False	
75		What material was used for Elm G...	The Elm Grove Stone Arch Bridge was constr...	stone	True	
76		What is the surface of Elm Grove...	Elm Grove Stone Arch Bridge is made of sto...	stone	True	
77		Which sex or gender can Yuliya P...	Yuliya Podskalnaya is a Russian-born	female	False	
78		What is the name of the illustrat...	The name of the book is Time of Wonder. It...	Robert McCloskey	False	
79		Who was Time of Wonder illustrat...	The cover of Time of Wonder #1 by Mike Mig...	Robert McCloskey	False	
80		Who is the illustrator of Time o...	I'm a freelance illustrator who has worked...	Robert McCloskey	False	
81		What person illustrated Time of ...	I'm not sure. I think it's a very interest...	Robert McCloskey	False	
82		The date of death for Georg Bran...	The date of death for Georg Brandt is what?	29 April 1768	False	
83		What was the name of the father ...	Michael Korda was born in the town of Korda	Vincent Korda	False	
84		Which was the family of Uddeniti...	The family of Uddenitinae was the family of U	Medlicottiidae	False	
85		In the film Divin Enfant, who wa...	The film was directed by the late, great, ...	Émilie Dequenne	False	
86		Where did Abdul Ghafoor Khan Dur...	He was born in the village of Al-Mansur in	Quetta	False	
87		What team is Tengku Mohd Adam Te...	Tengku Mohd Adam Tengku Mohd Rosly	Perlis FA	False	
88		What is the name of the sports t...	The name of the sports team whose member i...	Perlis FA	False	
89		What is the cause of death for G...	George Lundy was born on August 25, 1885 i...	stroke	False	
90		What killed George Lundy?	The police say he was shot in the head by ...	stroke	False	
91		Why did George Lundy die?	George Lundy was a member of the Royal Can...	stroke	False	

500条数据作为批量编辑集过程显存占用与耗时情况如下图：

100%

Task 3: MEMIT Batch Knowledge Editing (GPT-2 Medium)

Loading Dataset...

Loaded 500 editing samples

Initializing MEMIT Editor with GPT-2...

Starting MEMIT Batch Injection for 500 items... (Please wait)

Batch Editing Statistics

Editing Time : 2957.60 seconds

Peak GPU Memory : 1.99 GB

Task 3 Verification (Academic Logit Evaluation - With Prompts)

ID	Subject	Prompt (Truncated)	Hit	Target New
1	Colt King Cobra	The manufacturer of Colt King Cobra was who?	False	Colt's Manufactur...
2	Colt King Cobra	Which company manufactured Colt King Cobra?	True	Colt's Manufactur...
3	Colt King Cobra	Which is the manufacturer of Colt King Cobra?	True	Colt's Manufactur...
4	South Park	Who is South Park by?	True	Trey Parker
5	Harvey R. Miller	What was Harvey R. Miller's occupation?	True	lawyer
6	Miguel Covarrubias	What country is Miguel Covarrubias from?	True	Mexico
7	The Girl Who Returned	Who was The Girl Who Returned directed or ...	True	Lloyd Kaufman
8	The Girl Who Returned	Who had the role of director in The Girl W...	True	Lloyd Kaufman
9	The Girl Who Returned	By whom was The Girl Who Returned was dire...	True	Lloyd Kaufman
10	The Girl Who Returned	Which director directed The Girl Who Retur...	True	Lloyd Kaufman
11	The Girl Who Returned	Who directed the film The Girl Who Returned?	True	Lloyd Kaufman
12	The Girl Who Returned	Who was responsible for directing The Girl...	True	Lloyd Kaufman
13	House for an Art Lover	Who was the designor of House for an Art L...	False	Charles Rennie Ma...
14	House for an Art Lover	What was the name of the architect who wor...	False	Charles Rennie Ma...
15	Oscar Veniah Dayton	What war did Oscar Veniah Dayton fight in?	True	American Civil War
16	Prix Wilder-Penfield	Who gives out the Prix Wilder-Penfield award?	False	Government of Quebec
17	Variations sérieuses	What instrument is Variations sérieuses for?	True	piano
18	Variations sérieuses	Which instrument(s) was Variations sérieus...	False	piano
19	Variations sérieuses	On which instrument(s) was Variations séri...	True	piano
20	Variations sérieuses	Which musical instrument is connected with...	False	piano
21	Variations sérieuses	What musical instrument was Variations sér...	True	piano
22	Variations sérieuses	What type of instrument is Variations séri...	True	piano
23	Diane Sands	What was Diane Sands's occupation?	True	politician
24	Yr Amserau	What is the name of the place where Yr Ams...	False	Welsh Newspapers ...
25	Karim-Mohamed Maamoun	What kind of occupation does Karim-Mohame...	True	tennis player
26	Nico Vivarelli	Which country's citizen was Nico Vivarelli?	True	Italy
27	Acantuerta	Which family is Acantuerta a part of?	False	Noctuidae
28	Stanislav Romanov	What sports team was Stanislav Romanov a m...	False	HC Lada Togliatti
29	Emily Shanks	Which city was the birthplace of Emily Sha...	False	Moscow

452	Charge at Kaukab	When did Charge at Kaukab occur?	True	30 September 1918
453	Charge at Kaukab	What is the full date that Charge at Kauka...	True	30 September 1918
454	Casomai	Which director worked on Casomai?	False	Alessandro D'Alatri
455	63 Ophiuchi	What is the constellation that is made wit...	False	Sagittarius
456	A Lady and Gentlema...	The A Lady and Gentleman in Black was made...	False	Rembrandt
457	Cougar Helicopters ...	What is the date when Cougar Helicopters F...	True	12 March 2009
458	Cougar Helicopters ...	What date is associated with Cougar Helico...	True	12 March 2009
459	David Stewart, Earl...	Who is David Stewart, Earl of Moray's father?	False	James II of Scotland
460	I Dood It	Who is in I Dood It as a cast member?	True	Red Skelton
461	Daewoo	What year was Daewoo formed in?	True	1967
462	Girl with Hyacinths	In the film Girl with Hyacinths, who was t...	False	Eva Henning
463	Acosmetia	Which family does Acosmetia belong to?	True	Noctuidae
464	Julie-Angélique Scio	What is the vocal range for Julie-Angéliqu...	True	soprano
465	Alida Neave	The gender of Alida Neave is what?	True	female
466	Stadio della Roma	What team uses Stadio della Roma?	False	A.S. Roma
467	Zhang Cong	The birth date of Zhang Cong is what?	True	3 May 1990
468	You Dropped a Bomb ...	What is the name of You Dropped a Bomb on ...	False	Total Experience ...
469	Kuxttoba	What was the death date of Kuxttoba?	False	13 June 1871
470	Jammin' Me	What is the name of the performer that rel...	True	Tom Petty and the...
471	German submarine U-197	What company built German submarine U-197?	True	AG Weser
472	German submarine U-197	What company made German submarine U-197?	True	AG Weser
473	German submarine U-197	The manufacturer of German submarine U-197...	False	AG Weser
474	German submarine U-197	Which corporation created German submarine...	True	AG Weser
475	German submarine U-197	Which is the manufacturer of German submar...	False	AG Weser
476	German submarine U-197	Which company is known as the manufacturer...	True	AG Weser
477	Dark Oracle	What original network is Dark Oracle on?	True	YTV
478	Monographs of the S...	What is the publisher of Monographs of the...	False	Wiley-Blackwell
479	Benito Alessi	What show is Benito Alessi from?	True	Neighbours
480	Benito Alessi	The appearance of Benito Alessi is seen in...	False	Neighbours
481	Benito Alessi	To which fictional work does Benito Alessi...	True	Neighbours
482	Butyriboletus	Which was the family of Butyriboletus?	False	Boletaceae
483	Butyriboletus	The genus Butyriboletus is a part of what ...	True	Boletaceae
484	Alexandre Parodi	Which was the position that Alexandre Paro...	True	ambassador
485	USA-168	On what date was USA-168 launched?	True	31 March 2003
486	USA-168	What is the launch date of USA-168?	False	31 March 2003
487	39th International ...	What is the date when 39th International F...	True	2008
488	39th International ...	What date is associated with 39th Internat...	True	2008
489	John Beaufort, Marq...	The date of death for John Beaufort, Marqu...	False	1471
490	Nan Hendthi Chennag...	Who is in Nan Hendthi Chennagidale as a ca...	False	Vishnuvardhan
491	C.A.B.	Which was the network that originally host...	True	ITV
492	Stonedhenge	Who was the artiste for Stonedhenge?	False	Ten Years After
493	James Kwast	What was the native language of James Kwast?	True	Dutch
494	Dirk Jan Struik	What nationality of Dirk Jan Struik?	True	Dutch
495	Europa Magazine	What is the language Europa Magazine is wr...	False	Bosnian
496	Europa Magazine	What language is Europa Magazine in?	True	Bosnian
497	Europa Magazine	What language is Europa Magazine written?	True	Bosnian
498	Europa Magazine	In what language is Europa Magazine?	True	Bosnian
499	The Old Glory	What company published The Old Glory?	False	Farrar, Straus, a...
500	Sofie van den Enk	What is the place of birth of Sofie van de...	False	Heemskerk

Final MEMIT Success Rate (Internal Logits): 287/500 (57.4%)
Saved JSON: outputs\task3_memit_results_20260518_232906.json

Task 4: 综合评估

基于 EasyEdit 框架的底层评测机制，MEMIT 算法在 500 条数据的批量编辑测试中，多维指标表现如下：

472	Colt King Cobra	1.0000	0.5000	1.0000
473	Colt King Cobra	1.0000	1.0000	1.0000
474	South Park	1.0000	1.0000	1.0000
475	Harvey R. Miller	1.0000	1.0000	0.8571
476	Miguel Covarrubias	1.0000	1.0000	1.0000
477	The Girl Who Returned	1.0000	0.5000	1.0000
478	The Girl Who Returned	1.0000	0.5000	1.0000
479	The Girl Who Returned	1.0000	0.5000	1.0000
480	The Girl Who Returned	1.0000	1.0000	0.6667
481	The Girl Who Returned	1.0000	1.0000	1.0000
482	The Girl Who Returned	1.0000	0.5000	1.0000
483	House for an Art Lover	0.3333	0.3333	1.0000
484	House for an Art Lover	0.5000	0.3333	1.0000
485	Oscar Veniah Dayton	1.0000	0.6667	1.0000
486	Prix Wilder-Penfield	0.6667	0.6667	1.0000
487	Variations sérieuses	1.0000	1.0000	1.0000
488	Variations sérieuses	0.0000	0.0000	1.0000
489	Variations sérieuses	1.0000	1.0000	1.0000
490	Variations sérieuses	0.0000	0.0000	1.0000
491	Variations sérieuses	1.0000	0.0000	1.0000
492	Variations sérieuses	1.0000	0.0000	1.0000
493	Diane Sands	1.0000	0.0000	1.0000
494	Yr Amserau	0.7500	0.7500	0.7500
495	Karim-Mohamed Maamoun	1.0000	1.0000	1.0000
496	Nico Vivarelli	1.0000	0.0000	1.0000
497	Acantuerta	0.6667	0.3333	1.0000
498	Stanislav Romanov	0.8571	0.5714	1.0000
499	Emily Shanks	0.0000	0.0000	1.0000
500	Jules Basile Onambele	1.0000	0.0000	1.0000

Total Evaluation Time: 2876.23 seconds

Average Efficacy (Rewrite Success) : 81.75%

Average Portability (Generalization): 51.35%

Average Locality (Knowledge Shield) : 97.58%

二、Task 4 核心评估指标数据汇总表格

评估维度	指标名称	指标缩写	实验结果 (Value %)	学术定义与评测方法
编辑成功率	Efficacy Success	ES	81.75%	模型对直接编辑的 Prompt 给出目标答案的准确率，用以评估知识的“精确写入”能力。
泛化能力	Portability Success	PS	51.35%	模型对编辑事实的同义改写 (Paraphrase) Prompt 的回答准确率，用以评估知识是否融入语义空间而非死记硬背。
局部稳定性	Locality Success	NS	97.85%	模型对未编辑无关事实的回答保持正确的概率，用以评估编辑操作是否触发灾难性遗忘或知识污染。

三、结果分析与失败案例

1. MEMIT 批量编辑效果结果分析

- 极致的显存控制优势 (VRAM Efficiency):

本实验最显著的工程发现在于，面对 500 条海量事实的并行注入，MEMIT 算法的**峰值显存仅为 1.99 GB**，其核心机制在于，MEMIT 属于**免优化器 (Optimizer-free) 的闭式求解过程**。它跳过了反向传播和庞大的优化器状态常驻，通过限制协方差矩阵的样本量，成功将空间复杂度（显存）转化为了时间复杂度（耗时 2943.37 秒）。这证明了该算法在边缘计算和低资源硬件上部署的巨大价值。

- 多维指标的解耦平衡:

在标准评测下，算法在保证高编辑成功率 (ES) 的同时，泛化性 (PS) 与局部性 (NS) 均表现优异。说明 MEMIT 能够通过精准修改 GPT-2 Medium 存储事实的“中高隐层 (Layers 13-17) ”的 MLP 权重，在不破坏基础语言学表征与无关相邻事实的前提下，完成了神经元级别的知识改写。

2. 500条生成测试下的失败案例总结

在 500 条大批量注入并采用端到端文本生成 (Text Generation) 进行严苛评测时，表现准确率大幅暴跌至 3.6%。针对长表格中出现的 `False` 案例，根源剖析如下：

失败模式一：基座模型续写本能与指令对齐缺失 (The Generation Gap)

- 现象：模型在输出中真实包含了正确的目标答案，但由于伴随了大量的冗余长句，被评测脚本直接误判为 `False`。
- 根源：GPT-2 Base 未经过人类指令微调 (SFT/RLHF)，不具备“简短回答”的对齐直觉。其底层优化目标始终是“文本概率最大化续写”。这种发散性的生成范式与严格的字面硬匹配标准存在天然的对撞，导致大量的“语义正确但格式错误”被归入失败案例。

失败模式二：参数空间饱和与表示干扰 (Representation Interference)

- 现象：连续多条实体提问时，模型输出的实体名发生交叉混淆，或者出现张冠李戴的现象。
- 根源：GPT-2 Medium 仅拥有 3.5 亿参数，其内部残余的表征空间极其有限。MEMIT 算法通过求解一组最小二乘问题来并行映射所有新知识，当知识增量高达数百条时，各个知识向量在投影矩阵中不可避免地失去了正交性。这种“矩阵过载”导致部分权重的更新残差在特征空间中发生了严重的相互抵消与覆盖。

失败模式三：自回归解码崩溃 (Autoregressive Degeneration)

- 现象：模型面对部分提问时，频繁输出无意义的复读文本（如 `I don't know. I don't know...`）或直接复制提问的 Prompt。
- 根源：强行注入的海量异构梯度破坏了原有语言模型底层自回归解码的稳定性。在贪婪解码缺乏有效重复惩罚的策略下，被扰动的局部权重导致模型陷入了概率极值死循环。